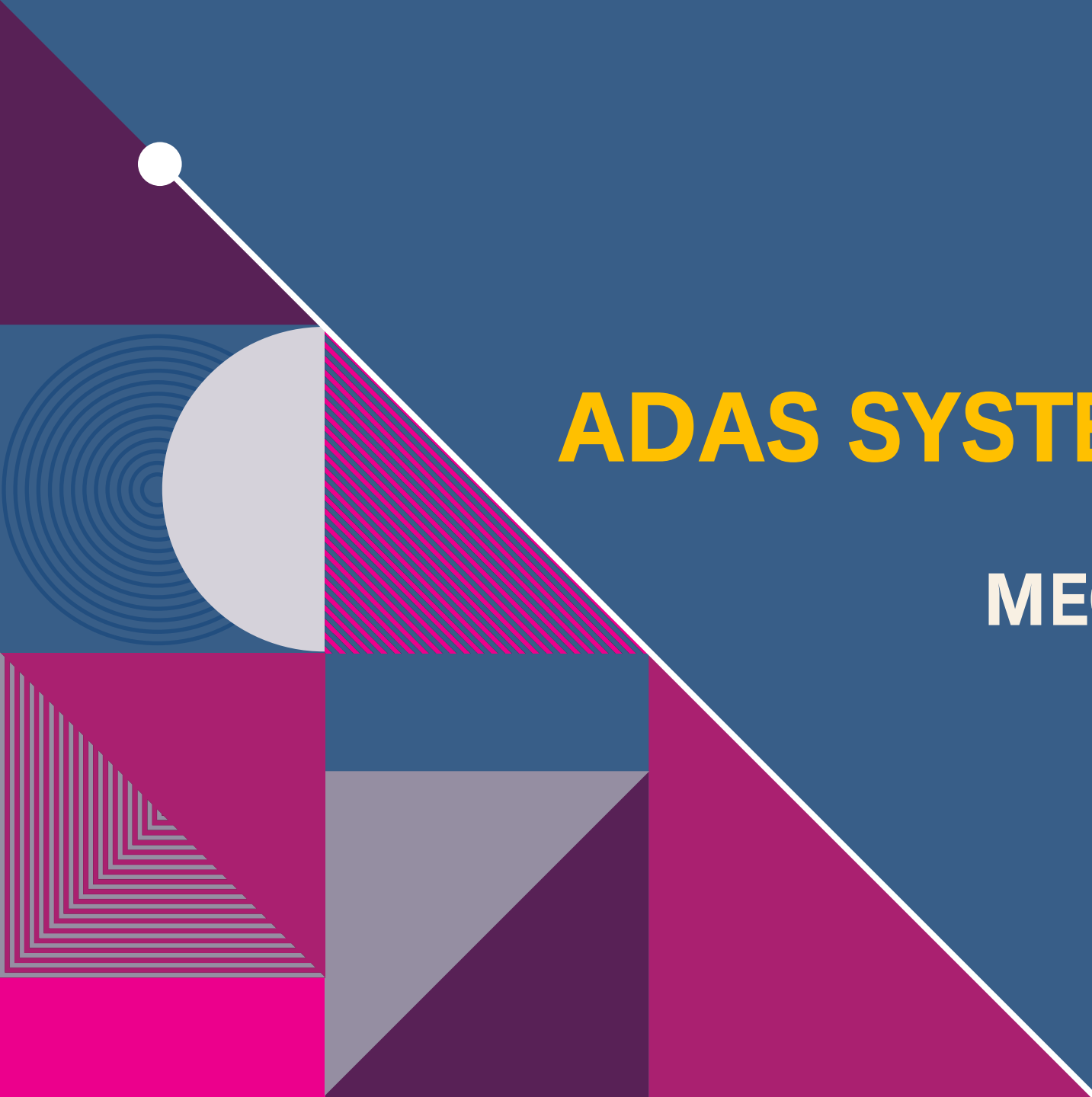
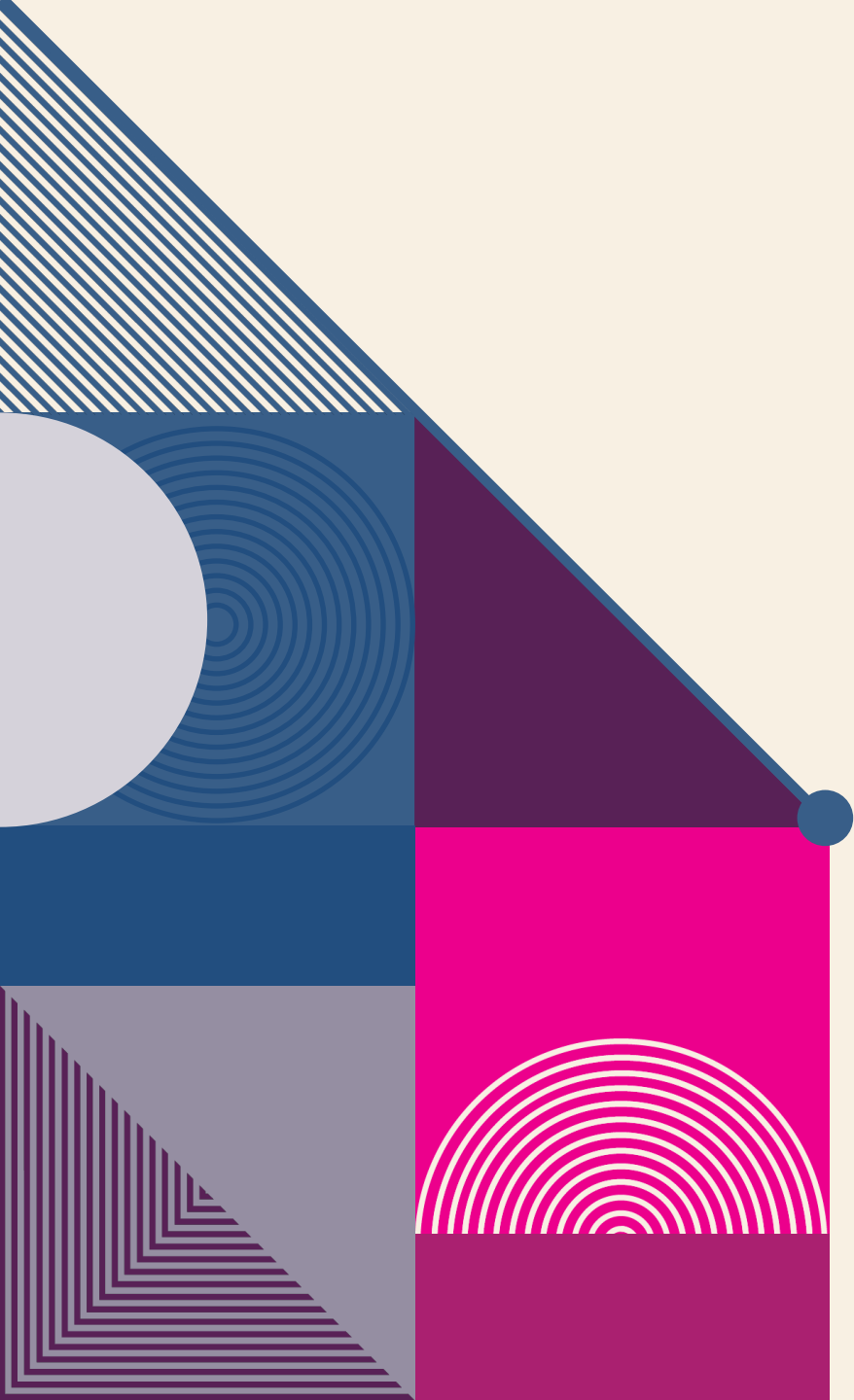




ADAS SYSTEM WITH CARLA

**MECHATRONICS SYSTEM
ENGINEERING**

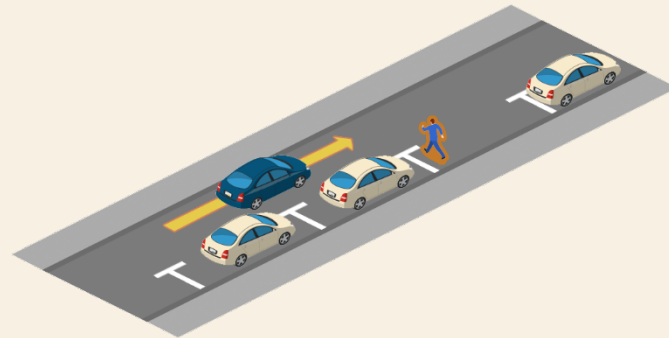
TEAM 3



AGENDA

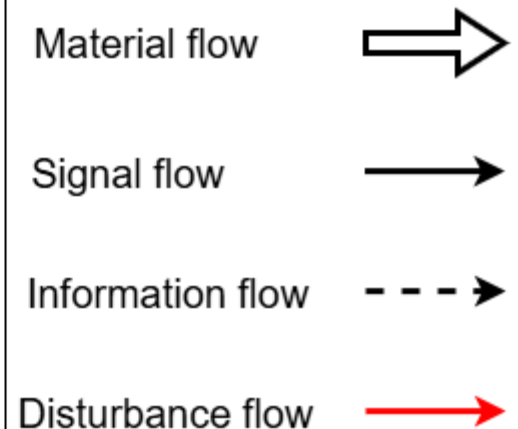
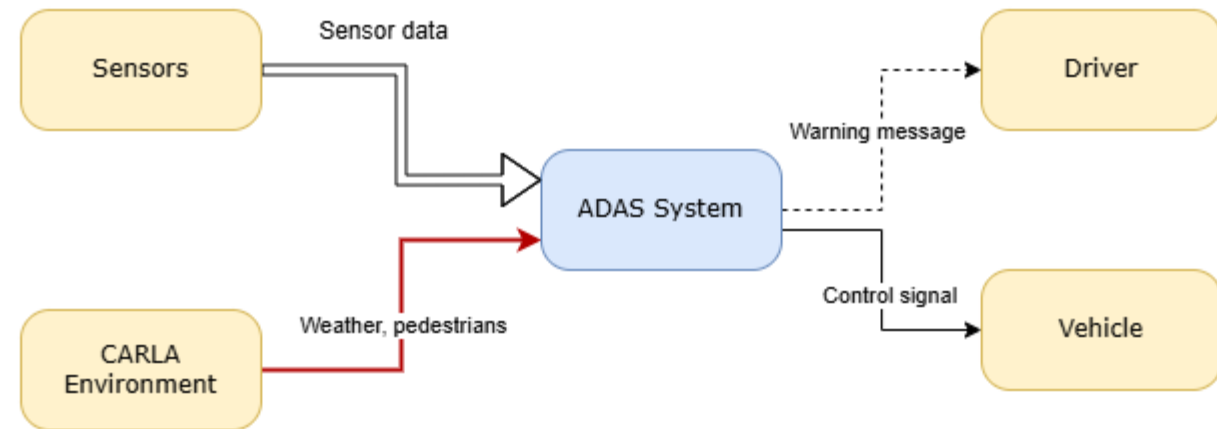
Introduction

SCENARIOS

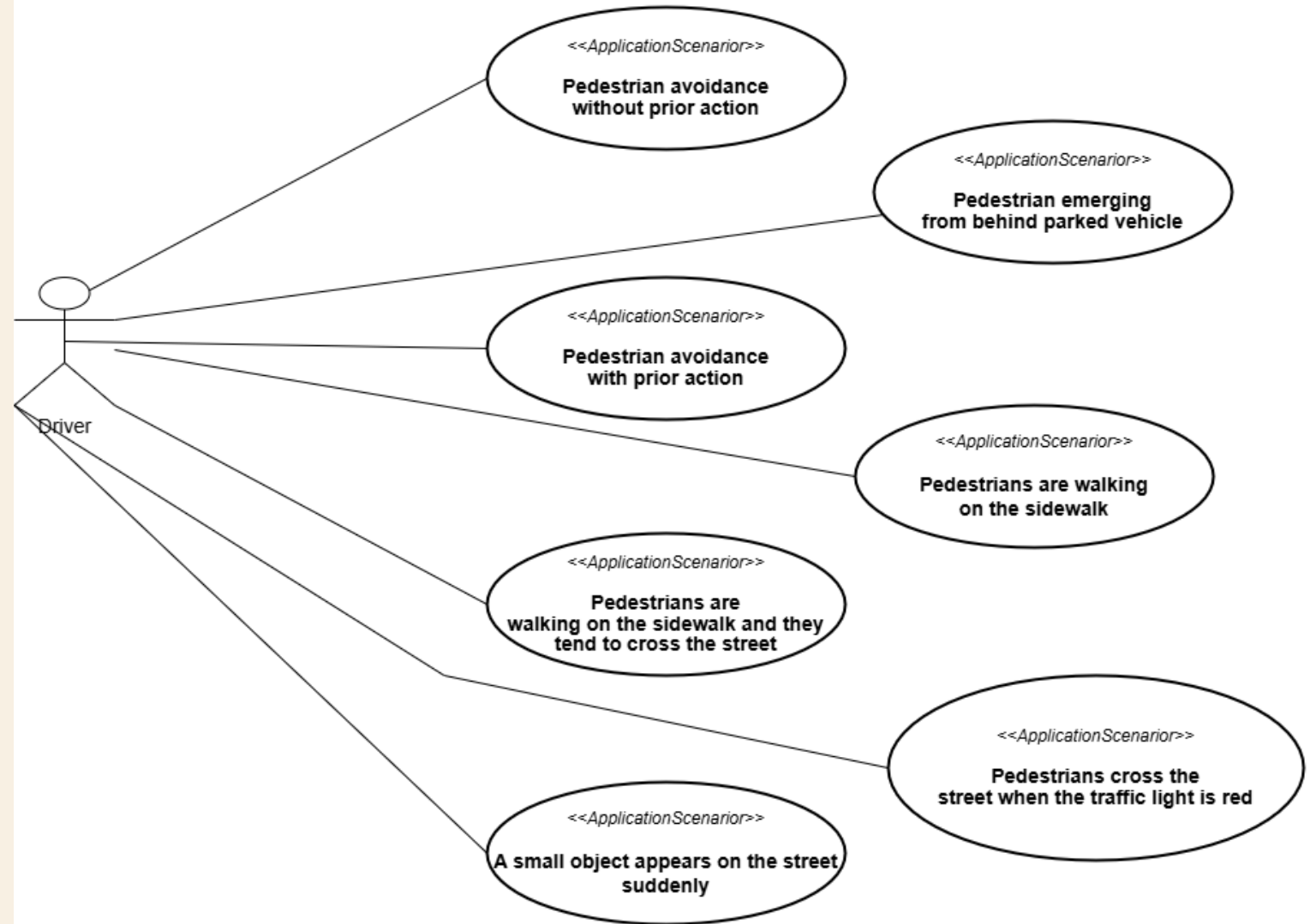


CONSENS – ENVIRONMENT MODEL

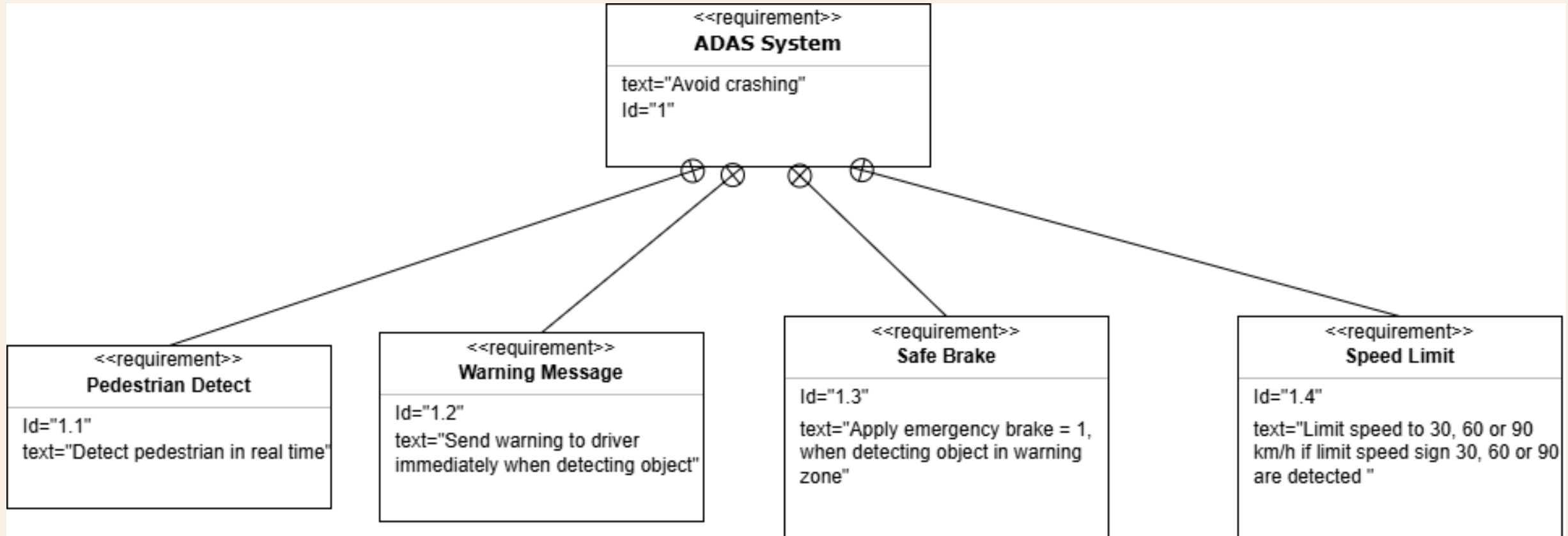
ibd [Package] Environment [Environment Diagram (IBD)]



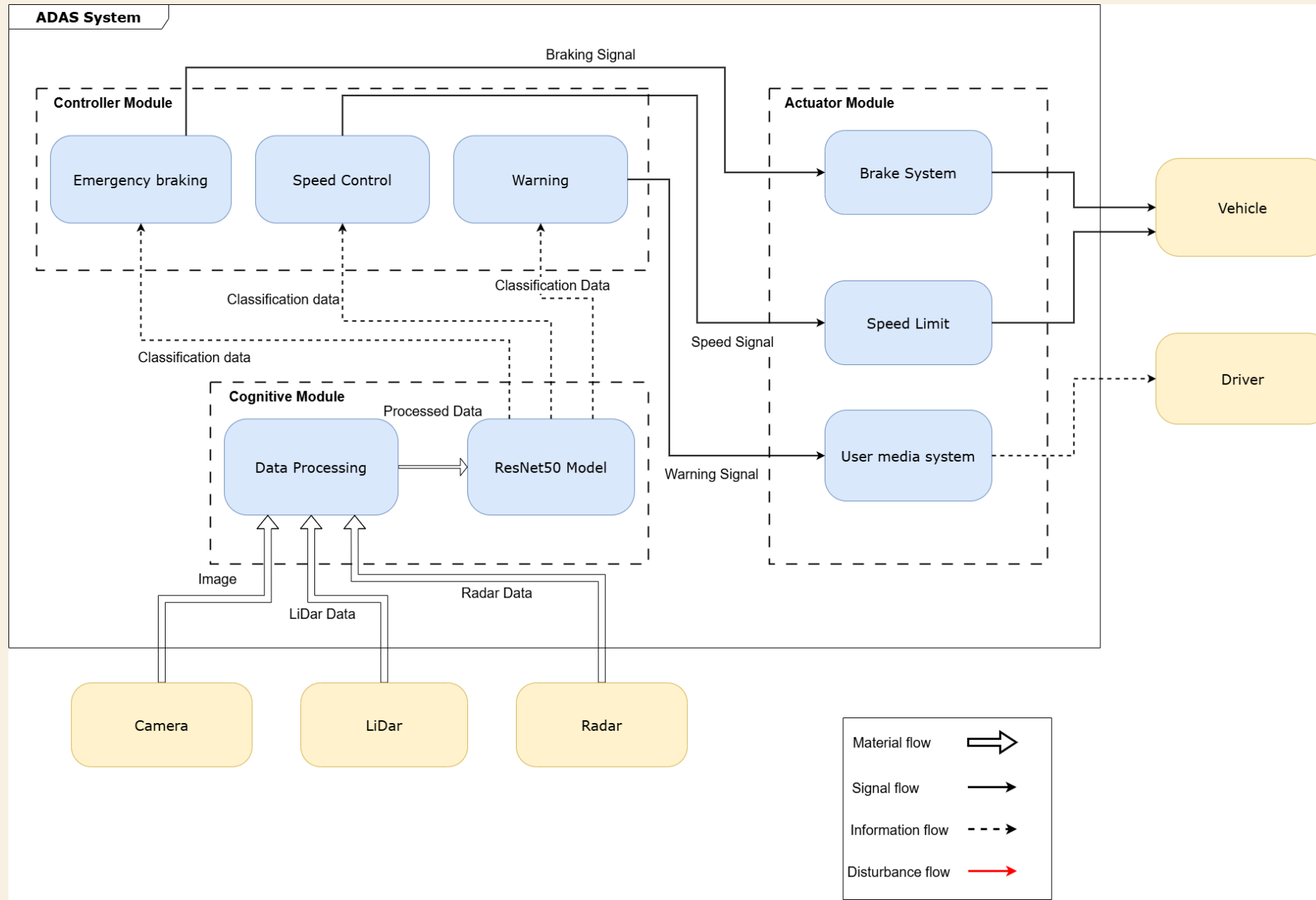
CONSENS – APPLICATION SCENARIOS



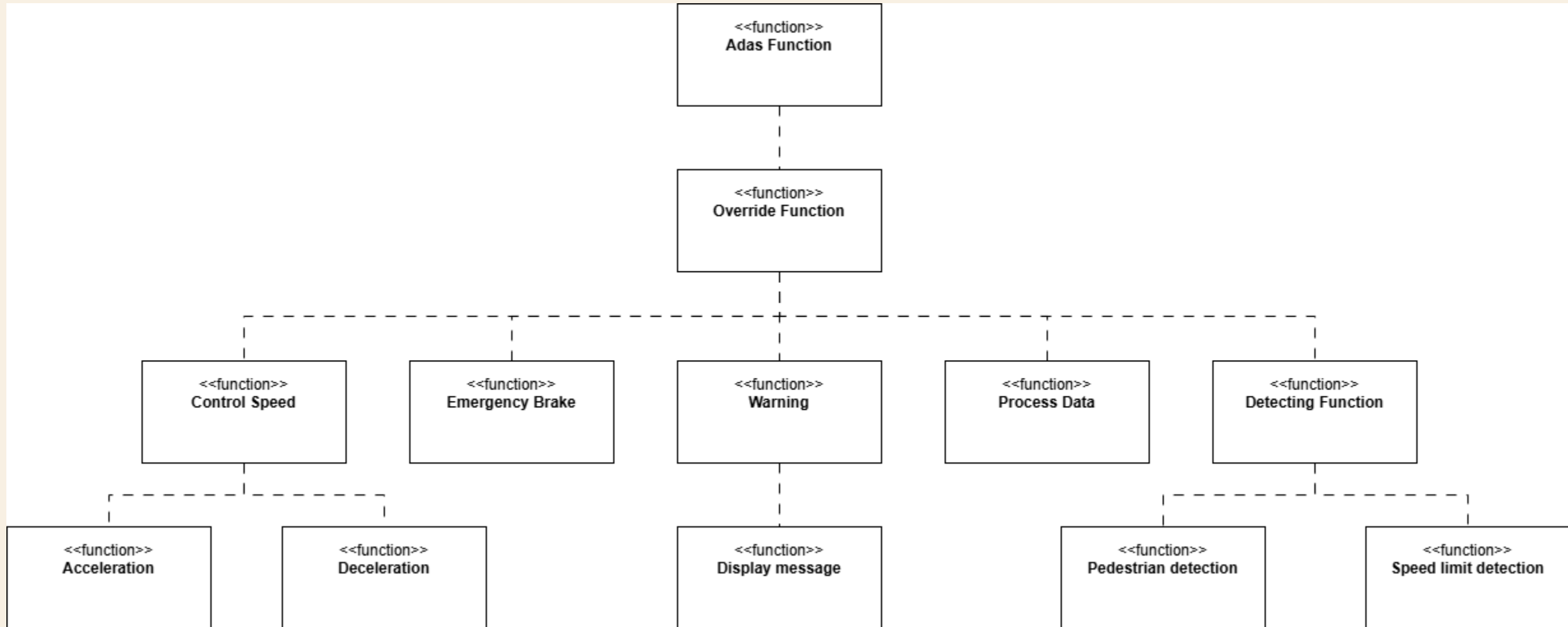
CONSENS – REQUIREMENT MODEL



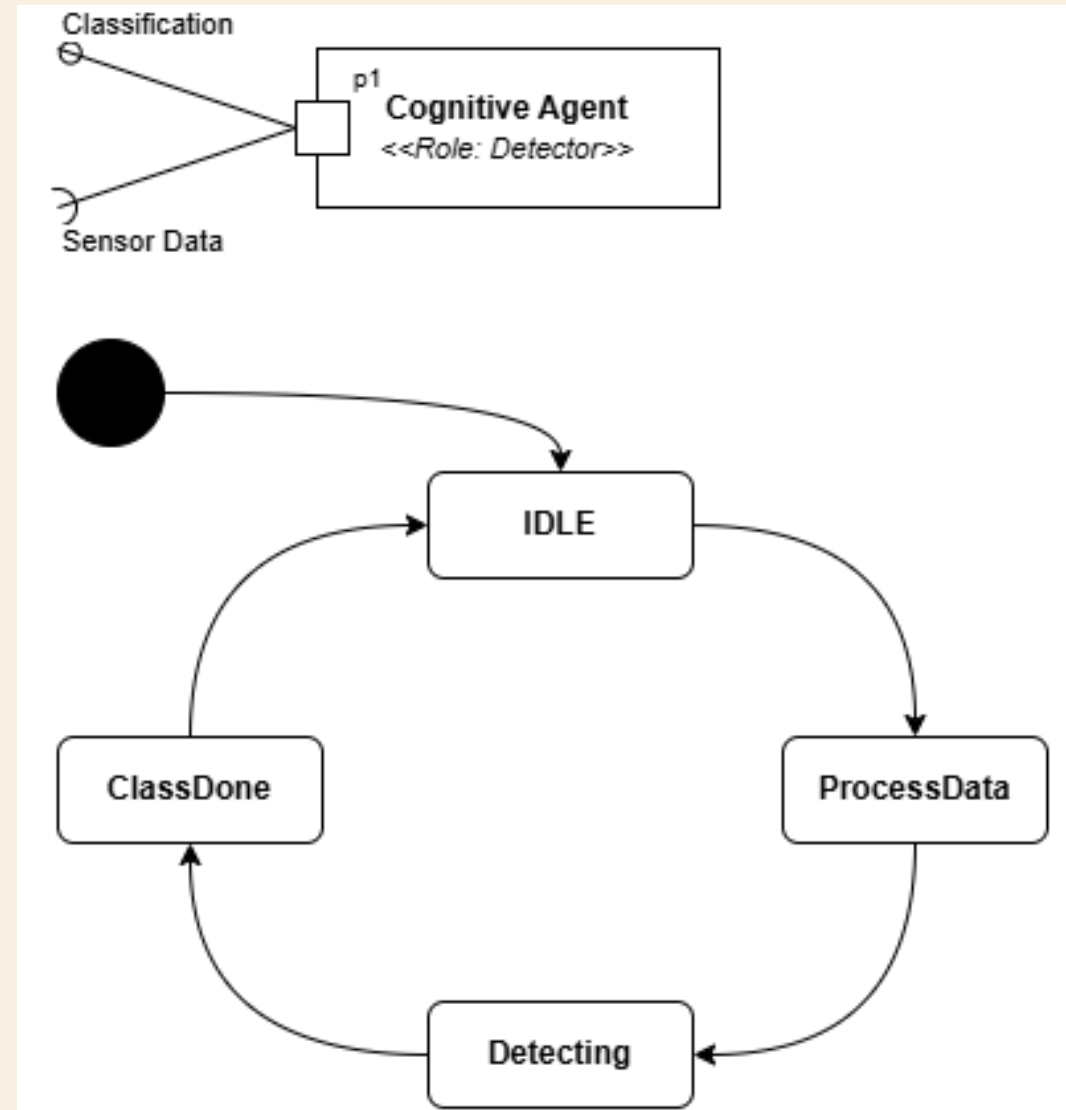
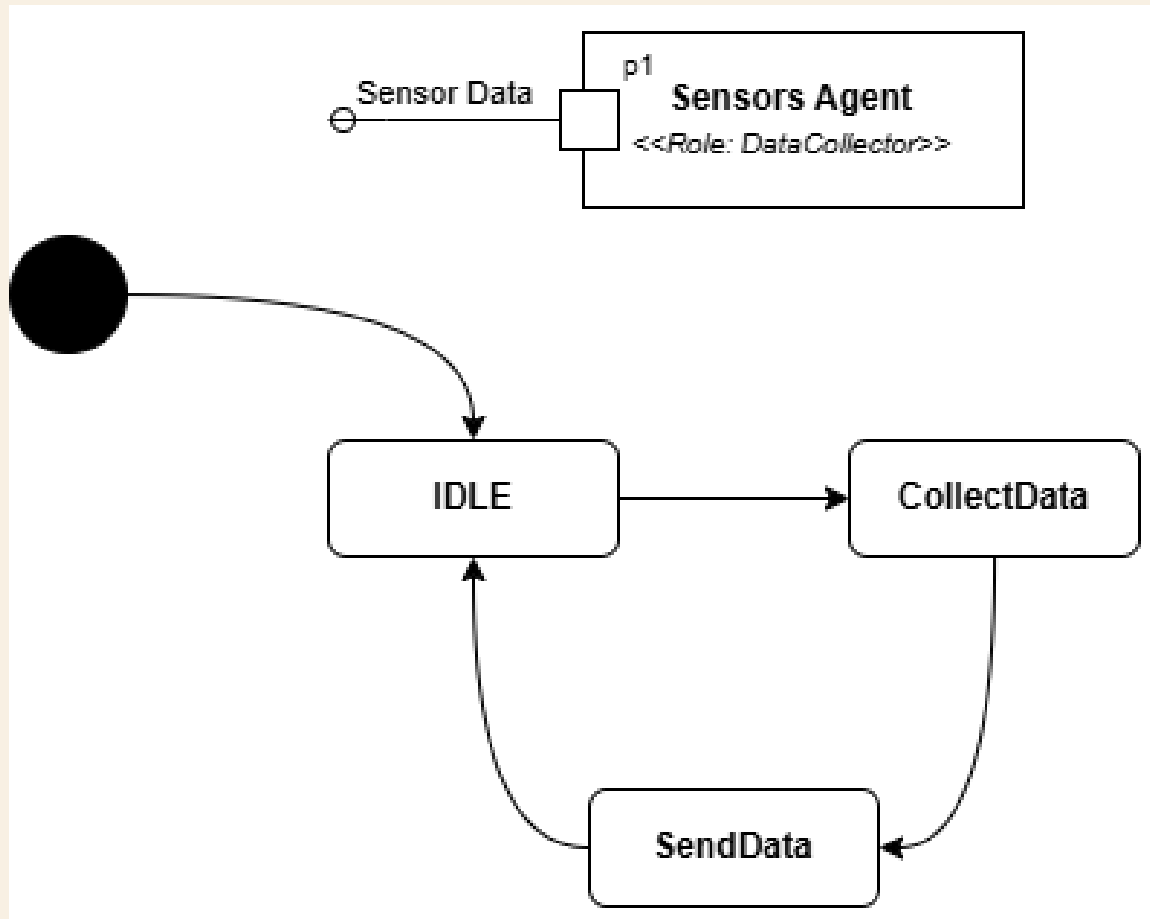
CONSENS – ACTIVE STRUCTURE



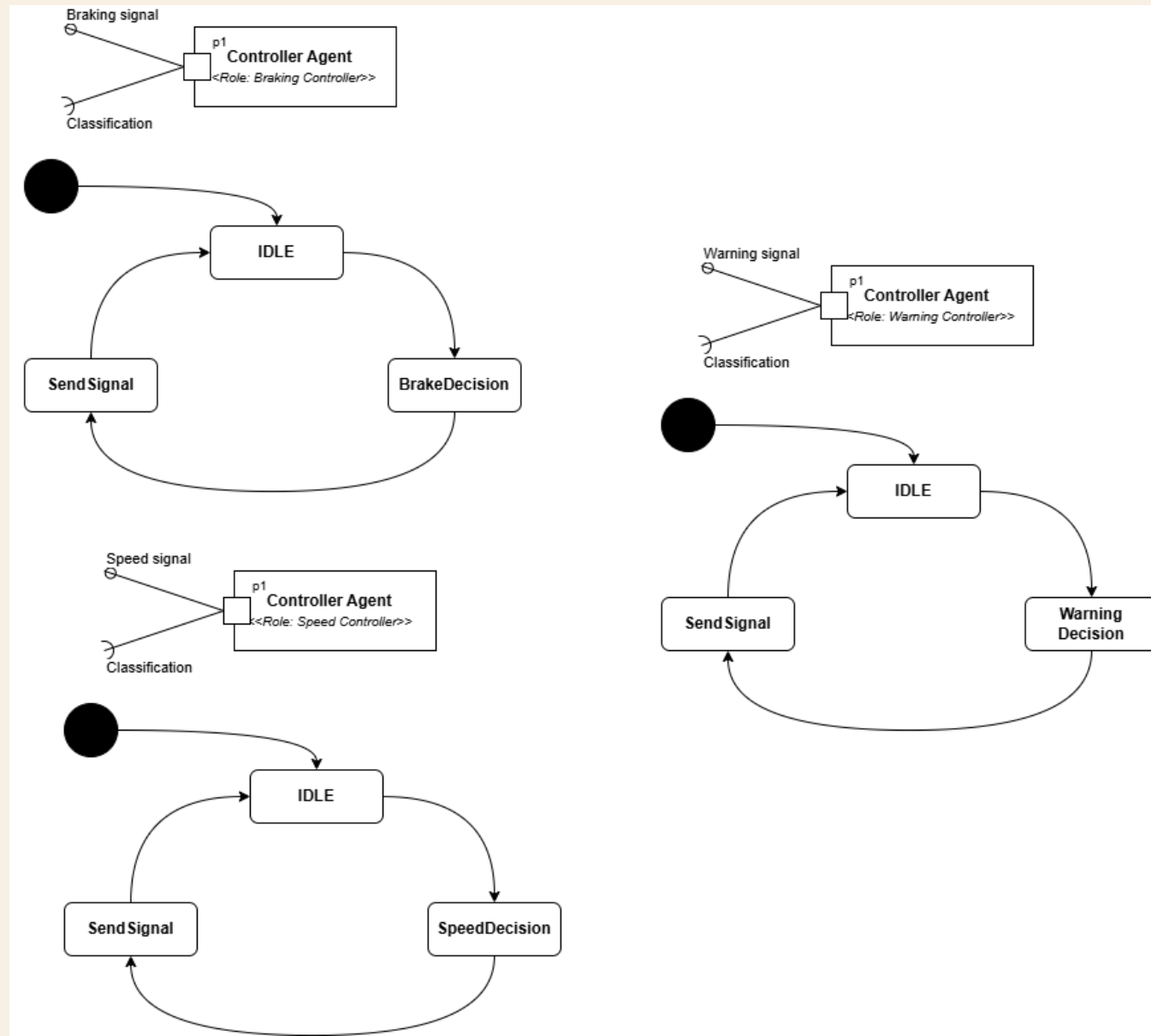
CONSENS – FUNCTION MODEL



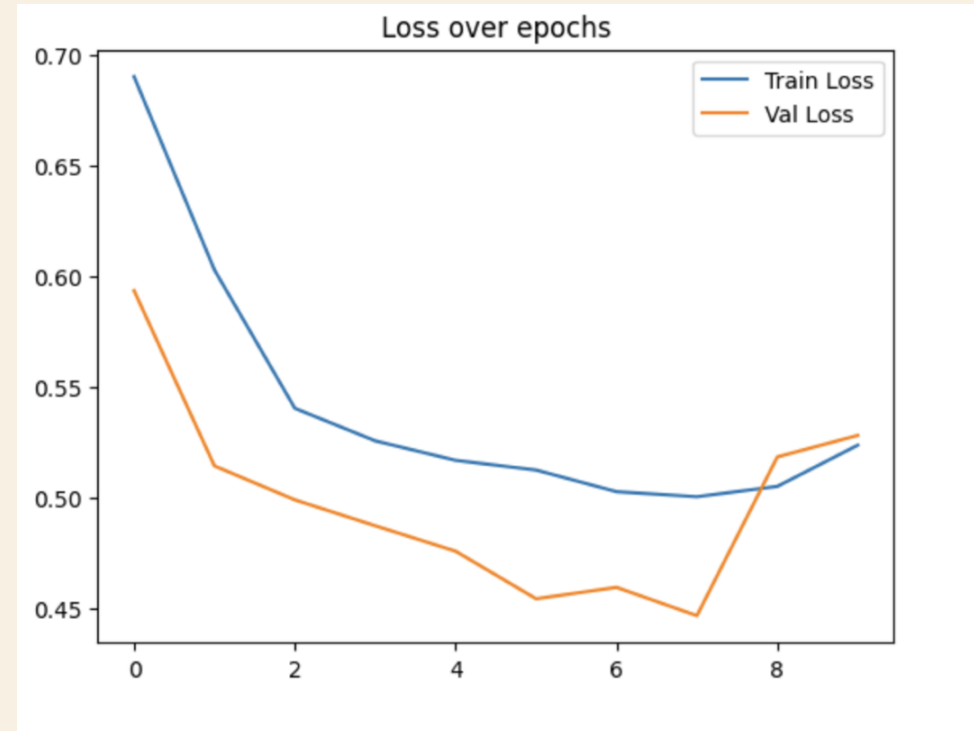
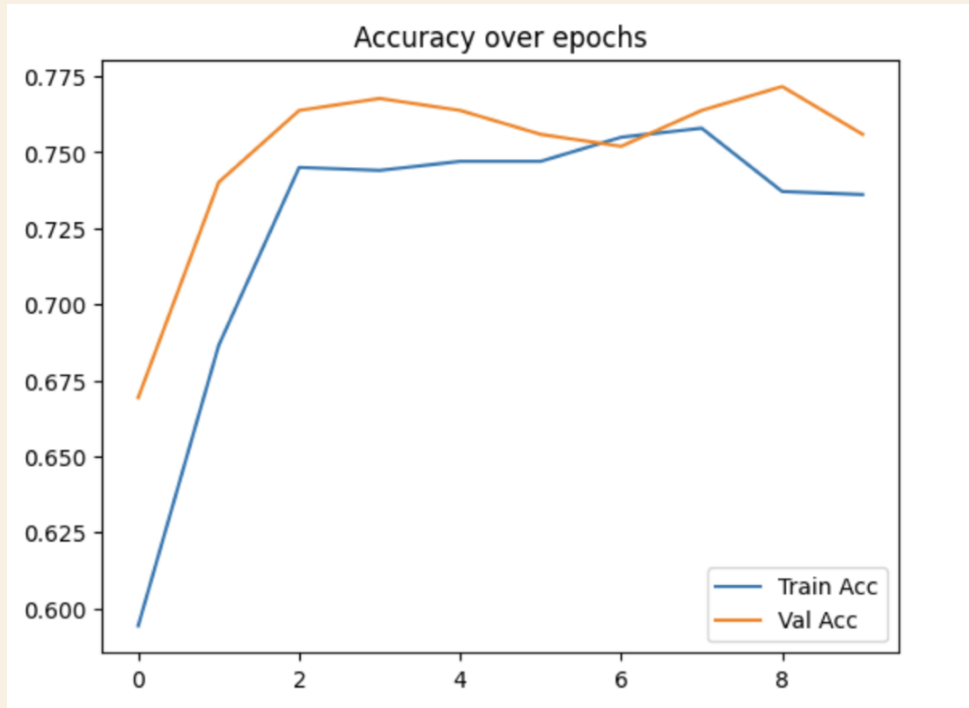
MECHATRONICS UML



MECHATRONICS UML



TRAINED MODEL RESULTS



TRAINED MODEL RESULTS

Model: "sequential_1"

Layer (type)	Output Shape	Param #
resnet50 (Functional)	(None, 7, 7, 2048)	23,587,712
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 2048)	0
dense_2 (Dense)	(None, 128)	262,272
dense_3 (Dense)	(None, 1)	129

Total params: 23,850,113 (90.98 MB)

Trainable params: 262,401 (1.00 MB)

Non-trainable params: 23,587,712 (89.98 MB)

/usr/local/lib/python3.11/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset` class

self._warn_if_super_not_called()

Epoch 1/10

32/32 ————— 43s 959ms/step - accuracy: 0.5489 - loss: 0.7345 - val_accuracy: 0.6693 - val_loss: 0.5935

Epoch 2/10

32/32 ————— 27s 608ms/step - accuracy: 0.6709 - loss: 0.6114 - val_accuracy: 0.7402 - val_loss: 0.5143

Epoch 3/10

32/32 ————— 23s 681ms/step - accuracy: 0.7349 - loss: 0.5541 - val_accuracy: 0.7638 - val_loss: 0.4990

Epoch 4/10

32/32 ————— 41s 701ms/step - accuracy: 0.7429 - loss: 0.5440 - val_accuracy: 0.7677 - val_loss: 0.4873

Epoch 5/10

32/32 ————— 20s 638ms/step - accuracy: 0.7581 - loss: 0.5030 - val_accuracy: 0.7638 - val_loss: 0.4758

Epoch 6/10

32/32 ————— 22s 701ms/step - accuracy: 0.7423 - loss: 0.5110 - val_accuracy: 0.7559 - val_loss: 0.4543

Epoch 7/10

32/32 ————— 20s 633ms/step - accuracy: 0.7391 - loss: 0.5174 - val_accuracy: 0.7520 - val_loss: 0.4595

Epoch 8/10

32/32 ————— 22s 691ms/step - accuracy: 0.7609 - loss: 0.4997 - val_accuracy: 0.7638 - val_loss: 0.4467

Epoch 9/10

32/32 ————— 20s 627ms/step - accuracy: 0.7335 - loss: 0.4994 - val_accuracy: 0.7717 - val_loss: 0.5184

Epoch 10/10

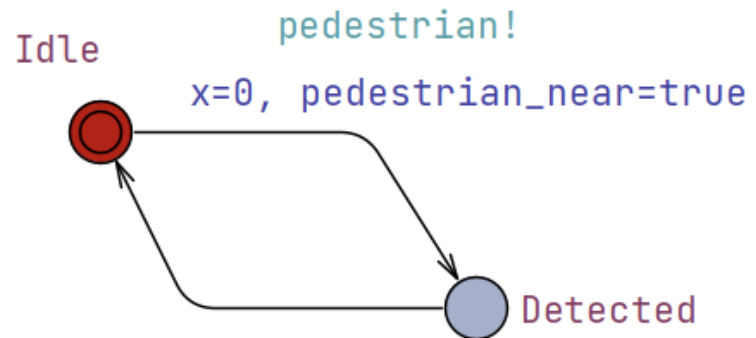
32/32 ————— 24s 764ms/step - accuracy: 0.7428 - loss: 0.5057 - val_accuracy: 0.7559 - val_loss: 0.5281

MODEL TESTING – REAL DATASET

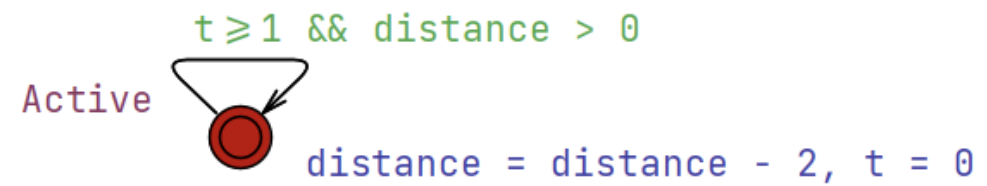


UPPAAL

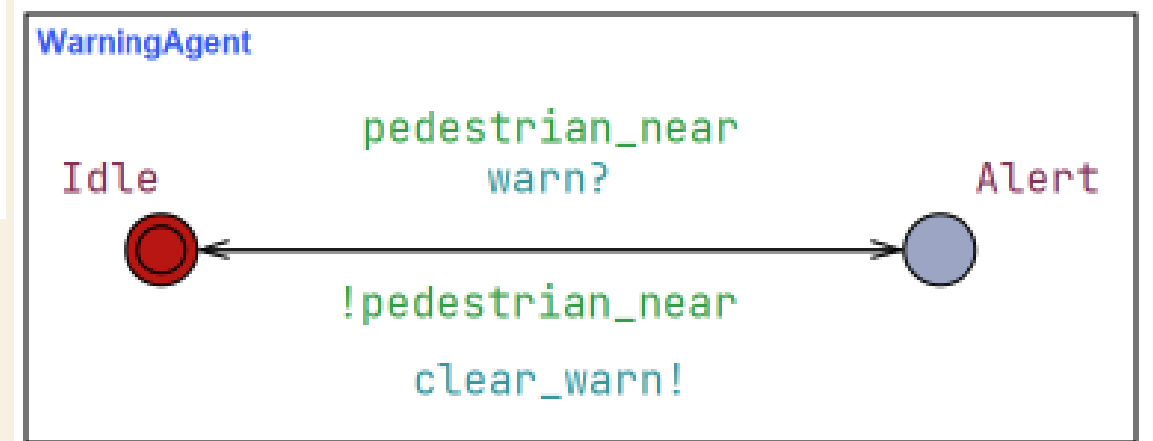
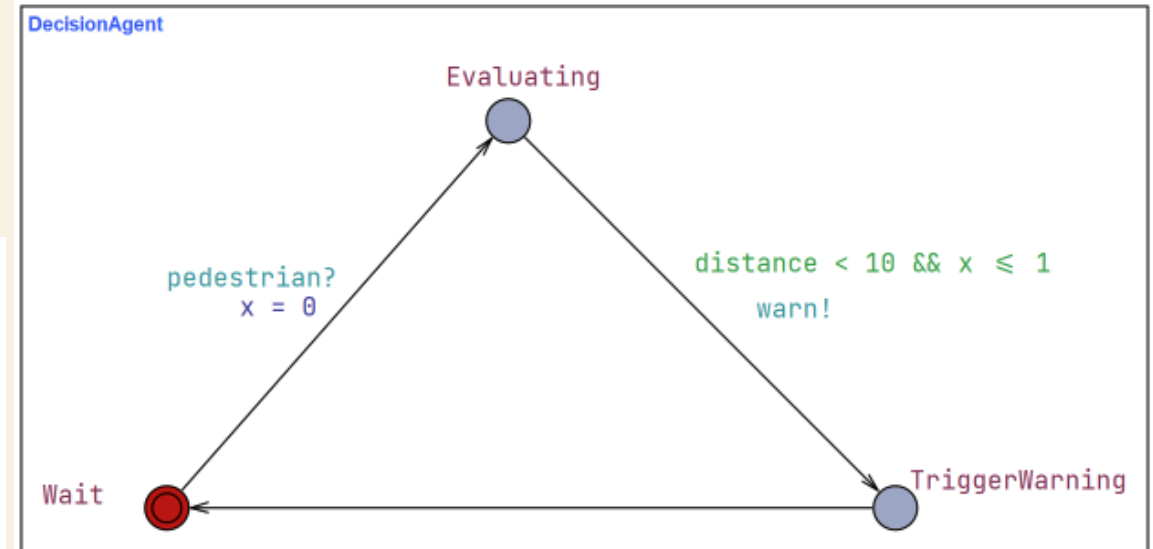
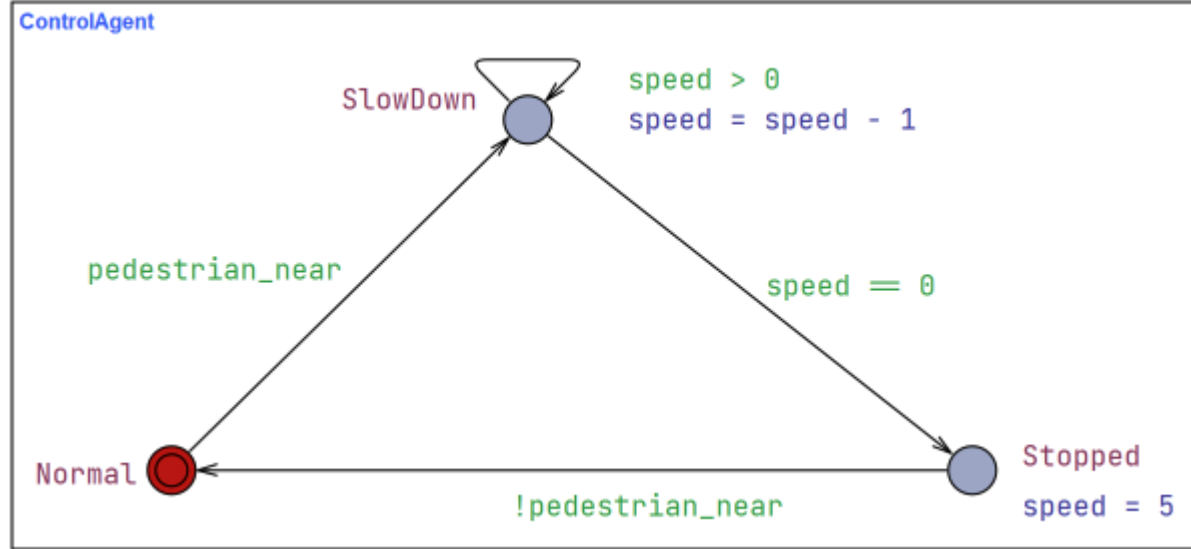
SensorCognitive



LIDAR



UPPAAL



UPPAAL

Functional Properties:

```
Status
Disconnected.
Established direct connection to local engine.
UPPAAL version 5.0.0 (rev. 714BA9DB36F49691), June 2023 -- server. Licensed to Phuc / FH Dortmund (stud.fh-dortmund.de)
A[] not deadlock
Verification/kernel/elapsed time used: 0.016s / 0s / 0.032s.
Resident/virtual memory usage peaks: 24,836KB / 74,996KB.
Property is satisfied.
```

=> Deadlock free

Safety Properties:

```
Status
Resident/virtual memory usage peaks: 24,884KB / 74,912KB.
Property is satisfied.
A[] speed >= 0
Verification/kernel/elapsed time used: 0s / 0s / 0.015s.
Resident/virtual memory usage peaks: 24,972KB / 75,072KB.
Property is satisfied.
```

=> Speed never goes negative

Safety Properties:

```
Status
Verification/kernel/elapsed time used: 0.015s / 0s / 0.016s.
Resident/virtual memory usage peaks: 25,124KB / 75,256KB.
Property is satisfied.
A[] (ControlAgent.Stopped imply speed == 0)
Verification/kernel/elapsed time used: 0s / 0s / 0.016s.
Resident/virtual memory usage peaks: 25,144KB / 75,288KB.
Property is satisfied.
```

=> The system never enters Stopped state unless speed is 0

An abstract geometric design on the left side of the slide. It features a dark blue background with various geometric shapes and patterns. A white circle is positioned near the top left. Below it, a light blue semi-circle is visible. To the right of the semi-circle, there is a pink triangle with diagonal lines. Below the semi-circle, there is a pink square with a pattern of concentric lines. To the right of the square, there is a light blue triangle. Below the square, there is a pink triangle. To the right of the triangle, there is a dark blue triangle. The overall design is modern and minimalist.

THANK YOU